# WEEK 7 QUESTION BANK

1. **Perform data exploration, preprocessing and splitting on Iris dataset from Scikit learn**

2. **Explain Evaluation Metrics in Model Evaluation and Testing ?**

3. **The Confusion matrix for machine learning model is given below, Evaluate Accuracy, Recall and Precision, Specificity F1-Score**

| | |
|---|---|
| 4. | 5. |
| 6. | 7. |

4. **Given a dataset containing 14 records, 9 belonging to class Yes and 5 belonging to class No, calculate the entropy of the dataset using the information gain approach.**

| Outlook | Temperature | Humidity | Wind | Played football(yes/no) |
|---|---|---|---|---|
| Sunny | Hot | High | Weak | No |
| Sunny | Hot | High | Strong | No |
| Overcast | Hot | High | Weak | Yes |
| Rain | Mild | High | Weak | Yes |
| Rain | Cool | Normal | Weak | Yes |
| Rain | Cool | Normal | Strong | No |
| Overcast | Cool | Normal | Strong | Yes |
| Sunny | Mild | High | Weak | No |
| Sunny | Cool | Normal | Weak | Yes |
| Rain | Mild | Normal | Weak | Yes |
| Sunny | Mild | Normal | Strong | Yes |
| Overcast | Mild | High | Strong | Yes |
| Overcast | Hot | Normal | Weak | Yes |
| Rain | Mild | High | Strong | No |

5. **Build decision tree-based model in python for Breast Cancer Wisconsin (diagnostic) dataset from sci-kit learn**

6. **Differentiate between pre-pruning and post-pruning.**

7. **Consider a dataset containing 10 student records, where 6 students passed and 4 students failed. Calculate the entropy of the dataset using the formula and explain the result.**

8. **Explain entropy and information gain with a suitable example.**

9. **Compare Decision Tree, KNN, and Naive Bayes classification models based on their working principles, advantages, limitations, and applications.**

**10. Differentiate between binary classification and multi-class classification with one suitable real-world example for each.**